

Monitoring Models Before the Labels Arrive: A Benchmark for Performance Estimation, Drift Detection and Retraining under Natural Drift and Label Delay

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Abstract

Deployed models are monitored long before their true labels arrive: a mortgage default is known up to two years after the loan is made. Yet drift detectors and label-free performance estimators are usually evaluated on artificially injected drift with labels available immediately. We present an open benchmark that replays five real, timestamped data sources (Freddie Mac mortgages, US Census ACS, CDC BRFSS, NYC taxi trips and TabReD) as streams in which a monitor sees each label only when it would have arrived in practice, including the *natural* label delay of mortgage defaults. The benchmark scores four tasks on eight experiments: estimating the performance of a new batch, estimating the performance of the partly labeled “in-flight” book, detecting degradation, and deciding when to retrain, the last scored as regret against a hindsight-optimal schedule. Three findings stand out. (1) Input-drift tests (PSI, Kolmogorov–Smirnov, MMD, domain classifiers) alarm in every period of the median experiment: on real data the inputs always change. (2) Label-free estimators miss shocks that change outcomes but not inputs (the 2007–08 mortgage crisis), while label-based monitoring sees them about two years late. (3) A fixed retraining calendar beats monitor-triggered retraining in most experiments. We also propose delay-adjusted CBPE, which uses early-arriving labels without being biased by them. It gives the best degradation alarm overall (median rank correlation with true degradation 0.58, precision 0.84). Its chain-ladder form fails, like actuarial chain-ladder methods, under calendar-time shocks such as COVID forbearance; an age–period–cohort hazard variant fixes this and gives the lowest in-flight error on Freddie Mac in three of four cases over five seeds, e.g. 31% lower default-rate error than the best label-based baseline across the 2007–2024 period. Code, data loaders and a leaderboard are released under the MIT license.

1 Introduction

Teams that run models in production ask two questions continually: *is the model still working*, and *should we retrain it*? Answering with labels is often impossible in time. A credit model’s outcome (default within 24 months) is fully known only two years after each decision; insurance conversions, medical outcomes and fraud chargebacks arrive weeks to months late. Monitoring therefore relies on proxies: tests that the input distribution has shifted [9, 17], estimators that predict accuracy from the model’s own confidence [8, 11, 15], and streaming detectors on whatever labels have arrived [1, 7].

These methods are rarely evaluated under the conditions they are meant for. Benchmarks typically inject synthetic drift into shuffled data and reveal each label immediately [3]; distribution-shift benchmarks with natural shift [12, 19] evaluate models, not monitors, and ignore when labels arrive. Delayed labels have been studied for stream classifiers [10], but not as the central obstacle to monitoring.

We build a benchmark around that obstacle. Our contributions:

1. **Streams with label arrival times.** Five public sources replayed in time order; each row carries the time it is scored and the time its label becomes known. For Freddie Mac mortgages the arrival is natural: a default is known in the month it happens, a repayment only when the 24-month window closes (Section 3).
2. **Four tasks with leakage-free evaluation:** next-batch estimation, in-flight book estimation, degradation detection, and retraining decisions scored against a hindsight-optimal schedule computed by dynamic programming (Section 4).
3. **Delay-adjusted CBPE**, a chain-ladder [14] correction that lets confidence-based estimation use early, biased labels, and an age-period-cohort hazard variant robust to calendar-time shocks (Section 5).
4. **An empirical study** of 11 detectors, 8 estimators and 9 retraining policies on 8 experiments (Section 6), with a public leaderboard.

2 Related work

Drift detection. Two-sample tests on inputs or model outputs [17], kernel tests [9], domain classifiers, and the population stability index used in credit risk flag that *data* changed. Streaming detectors such as ADWIN [1] monitor error rates and therefore need labels; see Gama et al. [7] for a survey.

Label-free performance estimation. ATC [8] and DoC [11] predict accuracy from confidence; CBPE [15] computes expected confusion matrices from calibrated probabilities. All are exact under pure covariate shift with calibration and fail under concept shift; label-shift estimators [13] assume $P(x | y)$ is fixed. None uses the labels that do arrive early.

Benchmarks with natural shift. Folktables [4] exposes year-to-year census shift; TabReD [18] supplies industrial tabular data with time-based splits; WILDS and Wild-Time [12, 19] cover natural shift for model robustness. Our benchmark reuses Folktables and TabReD data but evaluates monitors and adds label arrival.

Delayed outcomes in actuarial science. Insurers estimate claims incurred but not yet reported with chain-ladder development patterns [14]; competing-risk models [5] handle outcomes that can be pre-empted (a loan that repays cannot default). Our method borrows both ideas.

3 Benchmark

Stream format. A stream is a table of prediction events. Row i has features x_i , an event time t_i when the model scores it, an eventual label y_i (possibly never resolved) and a label time $\tau_i \geq t_i$ when y_i becomes known. At any decision time s , a monitor may use y_i only if $\tau_i \leq s$.

Table 1: Experiments. Delay: natural, or simulated constant delay.

Experiment	Target	Rows	Delay	Trained on	Monitored
freddie_2007	default $\leq 24m$	1.36M	natural	1999–2003	2007Q1–2024Q1 (q)
freddie_2016	default $\leq 24m$	1.36M	natural	2010–2012	2016Q1–2024Q1 (q)
acs_income_2016	income $> \$50k$	1.0M	365 d	2014	2016–2024 (y)
brfss_2014	diabetes	1.68M	180 d	2011–mid 2013	2014–2025 (q)
tlc_2019	tip $\geq 25\%$	1.8M	30 d	Feb–May 2019	Jul 2019–Dec 2021 (m)
tabred_homecredit	default	382k	90 d	Jan–May 2019	14 months (m)
tabred_homesite	conversion	261k	30 d	Jan–Aug 2013	20 months (m)
tabred_ecom	repeat purchase	160k	14 d	Mar 1–17 2013	5 weeks (w)

Data. Table 1 lists the streams. Freddie Mac [6]: 1.36M loans originated 1999–2026 (the public 50k-per-year sample); label: 90+ days delinquent or a credit-loss termination within 24 months of first payment; the label time is the month of the event, or the month the loan reaches age 24 or prepays. Age is counted in calendar months: the reported loan age is reset when a loan is modified, which would otherwise let defaults years later count as early ones. ACS [4]: income above \$50k, 2014–2024 (2020 has no standard release), relationship and state codes harmonized across years. BRFSS [2]: diagnosed diabetes, 2011–2024, ordered by interview month, 28 variables harmonized across renamings. NYC taxi [16]: card trips 2019–2021, label: tip $\geq 25\%$ of fare. TabReD [18]: the three binary tasks. Streams other than Freddie Mac receive a simulated constant delay (Table 1).

Training without label leakage. At the training date, a model may only learn from cohorts whose labels have (at least 95%) arrived. Using partly labeled cohorts is a common silent error: early-arriving labels (defaults) are over-represented. The last 12 months of trainable cohorts form the reference window on which monitors are calibrated. The model is a gradient-boosted tree ensemble; it is then frozen.

4 Tasks

T1: next-batch estimation. For each period, estimate the frozen model’s metric (AUC, positive rate, and accuracy where classes are balanced) on the batch scored in that period, using its scores and the labels arrived by the period’s end. Score: mean absolute error (MAE) over periods, with 95% moving-block bootstrap intervals over periods.

T2: in-flight book estimation. For long-horizon labels, a new batch’s outcome depends on events that have not happened yet: Freddie Mac loans made in 2019Q1 defaulted at 4.4% because of a pandemic a year later. The question a lender can answer is the state of its *book*: all loans scored in the last 24 months, part of which are already labeled. Same score as T1.

T3: degradation detection. A period is *degraded* if its true metric is worse than the reference by more than Δ (any change, for positive rate). Each detector returns a drift score and an alarm. Scores: Spearman correlation of the drift score with the true degradation; AUROC for several Δ ; precision and recall of the alarms.

T4: retrain or not. After each period a policy may retrain on the labels arrived so far. We precompute one candidate model per decision point and its loss (log loss) on every later period, so

any policy is evaluated by lookup. Cost = total loss + $\lambda \times$ retrains. We report regret against the hindsight-optimal schedule, found by dynamic programming over which model serves each period, for $\lambda = \rho \cdot g$, where g is the average gain of one retrain under “always retrain” and $\rho \in \{0, 0.25, 1, 4\}$.

5 Delay-adjusted CBPE

CBPE treats calibrated scores c_i as the probability each row is positive. Under concept shift the calibration breaks; the labels that would reveal this arrive late, and the first to arrive are biased: defaults arrive within months, repayments only after 24.

Arrival curves. From mature rows (older than the longest observed delay) we estimate $F(a)$, the probability that a positive’s label has arrived by age a , and $G(a)$ for negatives.

Offset. We model the true probability as $p_i = \sigma(\text{logit } c_i + \delta)$ and fit δ by maximum likelihood on the last 12 months of rows. Each row contributes p_i if its label arrived positive, $1 - p_i$ if negative, and

$$1 - p_i F(a_i) - (1 - p_i) G(a_i)$$

if still pending at age a_i . Early defaults therefore move δ at once without being over-counted: a pending row is evidence of a negative only to the extent that a positive would have arrived by now. With constant delay the method reduces to recalibrating on the most recent complete labels.

Estimates. For a new batch, the metric is computed as in CBPE from p_i (expected AUC from all positive–negative pairs; expected accuracy and positive rate in closed form). For the in-flight book, arrived labels count as observed and pending rows use the posterior $P(y_i = 1 \mid \text{pending}) = p_i(1 - F(a_i))/(1 - p_i F(a_i) - (1 - p_i)G(a_i))$.

Calendar-time shocks. The development pattern F is assumed stable. A shock that hits rows of every age at once violates it: during COVID forbearance, loans of all ages became 90 days delinquent within months, which the method reads as the early edge of a much larger wave and projects onto new loans. Actuarial chain-ladder methods share this weakness with “calendar-year effects”.

Hazard variant. We recast the method as an age–period–cohort model in discrete time. On a monthly age grid, row i ’s cumulative incidence without shocks is $p_i F(k)$, which defines monthly hazard increments $L_i(k) = \log(1 - p_i F(k - 1)) - \log(1 - p_i F(k))$. *Age* is fixed by F . *Cohort*: $p_i = \sigma(\text{logit } c_i + \delta_{g(i)})$ with one shift for the newest 12 months of rows and one for the 12 before. *Period*: in each of the last 24 calendar months t , every row alive gets the same extra hazard $\lambda_t \geq 0$. With $S_i(k) = \exp(-\sum_{j \leq k} [L_i(j) + \lambda_{t_i(j)}])$, a positive arriving in month k^* contributes $S_i(k^* - 1) - S_i(k^*)$, a known negative $S_i(H)$, and a pending row $S_i(a_i) - S_i(H) G(a_i)$. Older rows anchor λ : a calendar burst raises their arrivals too, a new-cohort shift does not. This is what separates the two. Two design choices matter. First, $\lambda_t \geq 0$: recent rows have no resolved negatives yet, so with signed period effects “nearly every row will be positive, just slowly” fits as well as the truth. Second, shocks are additive rather than proportional to each row’s hazard, since forbearance-type shocks hit safe and risky rows alike. New rows are projected with the newest cohort’s shift only; bursts are counted but not extrapolated. We fit by L-BFGS-B with analytic gradients.

Table 2: Degradation detection: medians over 16 cases.

Detector	Spearman ρ	Alarm rate	Precision	Recall
Delay-adjusted CBPE alarm (ours)	0.58	0.32	0.84	0.52
CBPE alarm	0.50	0.11	0.82	0.11
ADWIN on scores	0.50	0.10	0.67	0.13
KS on scores	0.49	1.00	0.48	1.00
PSI	0.40	1.00	0.44	1.00
MMD	0.38	1.00	0.52	1.00
Univariate KS / χ^2	0.37	1.00	0.48	1.00
Domain classifier	0.37	1.00	0.52	1.00
Latest complete cohort alarm	0.31	0.39	0.71	0.71
Recent arrivals alarm	0.21	0.24	0.63	0.40
ADWIN on arrived errors	0.17	0.13	0.83	0.19

6 Results

Detection (T3). Table 2 summarizes 16 cases (8 experiments $\times$ 2 event types). Every input-drift test alarms in every period of the median case: lender churn, interest-rate levels, a discontinued channel code and a field introduced in 2017 all shift Freddie Mac’s inputs without necessarily hurting the model. Their scores still rank severity moderately (median ρ 0.37–0.49), but their thresholds are unusable. The delay-adjusted alarm both ranks degradation best (median ρ 0.58) and alarms selectively (32% of periods, precision 0.84). ADWIN on arriving errors, the classic streaming setup, ranks last: under delay it sees too little, too late.

Estimation (T1, T2). Tables 3 and 4 report MAE over five seeds. Seed-to-seed variation is small relative to the gaps between methods.

Next batch. No method dominates. Label-free and delay-adjusted estimators are most accurate when shift is mostly in the inputs (TabReD homesite and homecredit, BRFSS). With short delays and a drifting label, simply using the latest complete labels wins (ACS income under wage inflation, NYC taxi tips, TabReD ecom). On short-horizon streams the hazard variant’s extra parameters are poorly anchored and cost accuracy (NYC taxi positive rate: 0.030 vs. 0.012 for the chain-ladder form), so it should be preferred only where labels arrive over a long horizon. On Freddie Mac the task is close to unanswerable: loans made in 2019Q1 defaulted at 4.4% because of a pandemic a year later.

In-flight book. Here early labels pay off (Table 4). The hazard variant has the lowest error in three of four cases. On *freddie_2007* its default-rate error is 0.0060 against 0.0087 for the best label-based baseline, and its AUC error 0.0180 against 0.0215; block-bootstrap intervals of the paired differences exclude zero (default rate: -0.0034 [-0.0076 , -0.0004] vs. waiting for complete cohorts; AUC: -0.0058 [-0.0126 , -0.0001] vs. labels to date). The exception is *freddie_2016* AUC, where labels to date is better (0.0263 vs. 0.0287; interval includes zero).

Figure 1 shows the mechanism. CBPE stays near the reference default rate through the 2007–08 crisis (loans made in 2008Q1: true 6.6%, CBPE 1.0%). Waiting for complete cohorts sees the crisis about two years late and then reports it after it has passed (4.7% for 2010Q1 loans against a true 0.6%). On the in-flight book both delay-adjusted estimators react within quarters. Under COVID, the chain-ladder form reads forbearance as the start of a much larger wave (peak in-flight estimate 8.3% against a true peak of 4.3%); the hazard variant attributes it to calendar months and peaks at 4.3%.

Table 3: Next-batch estimation. MAE, mean (std) over 5 seeds; lower is better, best in bold.

Experiment	Metric	Ref.	Recent	Complete	CBPE	DA-CBPE	DA-CBPE-HZ
acs_income_2016	accuracy	0.0369 (0.0018)	–	0.0087 (0.0002)	0.0314 (0.0024)	0.0145 (0.0020)	0.0154 (0.0021)
acs_income_2016	pos. rate	0.0929 (0.0015)	–	0.0199 (0.0002)	0.0778 (0.0022)	0.0178 (0.0006)	0.0202 (0.0014)
acs_income_2016	AUC	0.0096 (0.0020)	–	0.0019 (0.0003)	0.0071 (0.0021)	0.0091 (0.0022)	0.0090 (0.0022)
brfss_2014	pos. rate	0.0093 (0.0009)	0.0045 (0.0002)	0.0052 (0.0002)	0.0041 (0.0010)	0.0033 (0.0004)	0.0039 (0.0006)
brfss_2014	AUC	0.0106 (0.0007)	0.0042 (0.0003)	0.0062 (0.0003)	0.0050 (0.0001)	0.0051 (0.0001)	0.0051 (0.0001)
freddie_2007	pos. rate	0.0108 (0.0000)	0.0140 (0.0000)	0.0166 (0.0000)	0.0123 (0.0003)	0.0143 (0.0002)	0.0122 (0.0009)
freddie_2007	AUC	0.0498 (0.0034)	0.0502 (0.0033)	0.0471 (0.0025)	0.0382 (0.0075)	0.0360 (0.0063)	0.0375 (0.0069)
freddie_2016	pos. rate	0.0140 (0.0000)	0.0166 (0.0000)	0.0174 (0.0000)	0.0140 (0.0004)	0.0218 (0.0010)	0.0142 (0.0011)
freddie_2016	AUC	0.1054 (0.0063)	0.0724 (0.0077)	0.0422 (0.0029)	0.0726 (0.0044)	0.0691 (0.0038)	0.0713 (0.0039)
tabred_ecom	pos. rate	0.0697 (0.0019)	0.0772 (0.0000)	0.0966 (0.0000)	0.1741 (0.0094)	0.1449 (0.0075)	0.1717 (0.0213)
tabred_ecom	AUC	0.2316 (0.0255)	0.1014 (0.0128)	0.0717 (0.0154)	0.1133 (0.0288)	0.1089 (0.0294)	0.1083 (0.0292)
tabred_homecredit	pos. rate	0.0094 (0.0000)	0.0119 (0.0000)	0.0099 (0.0000)	0.0089 (0.0019)	0.0089 (0.0006)	0.0086 (0.0005)
tabred_homecredit	AUC	0.0180 (0.0027)	0.0254 (0.0026)	0.0185 (0.0034)	0.0159 (0.0018)	0.0159 (0.0017)	0.0159 (0.0017)
tabred_homesite	pos. rate	0.0236 (0.0016)	0.0160 (0.0000)	0.0160 (0.0000)	0.0052 (0.0013)	0.0046 (0.0015)	0.0051 (0.0017)
tabred_homesite	AUC	0.0137 (0.0017)	0.0058 (0.0001)	0.0056 (0.0003)	0.0051 (0.0020)	0.0048 (0.0020)	0.0049 (0.0021)
tlc_2019	accuracy	0.0062 (0.0005)	0.0036 (0.0001)	0.0031 (0.0000)	0.0055 (0.0004)	0.0086 (0.0003)	0.0144 (0.0005)
tlc_2019	pos. rate	0.0339 (0.0013)	0.0085 (0.0000)	0.0078 (0.0000)	0.0164 (0.0010)	0.0116 (0.0002)	0.0301 (0.0007)
tlc_2019	AUC	0.0189 (0.0009)	0.0060 (0.0002)	0.0052 (0.0001)	0.0057 (0.0002)	0.0078 (0.0003)	0.0073 (0.0001)

Table 4: In-flight book estimation (Freddie Mac, natural label delay). MAE, mean (std) over 5 seeds; lower is better, best in bold.

Experiment	Metric	Ref.	To date	Complete	CBPE	DA-CBPE	DA-CBPE-HZ
freddie_2007	pos. rate	0.0102 (0.0000)	0.0090 (0.0000)	0.0087 (0.0000)	0.0119 (0.0003)	0.0065 (0.0000)	0.0060 (0.0008)
freddie_2007	AUC	0.0462 (0.0035)	0.0215 (0.0009)	0.0394 (0.0016)	0.0333 (0.0086)	0.0220 (0.0046)	0.0180 (0.0021)
freddie_2016	pos. rate	0.0124 (0.0000)	0.0088 (0.0000)	0.0083 (0.0000)	0.0127 (0.0002)	0.0103 (0.0001)	0.0060 (0.0003)
freddie_2016	AUC	0.0968 (0.0058)	0.0263 (0.0012)	0.0355 (0.0018)	0.0692 (0.0037)	0.0422 (0.0023)	0.0287 (0.0024)

Synthetic checks. Two controlled streams isolate the two kinds of shock: a lasting shift (the outcome rate of new cohorts jumps) and a calendar burst (for four months every active loan has an extra 3% monthly default hazard, new cohorts unaffected). The chain-ladder form tracks the lasting shift best (in-flight MAE 0.007 vs. 0.050 for complete cohorts) but its error grows more than tenfold under the burst (0.094). The hazard variant gives up a little on the lasting shift (0.010) and cuts the burst error to 0.030, the best of all methods.

Retraining (T4). At moderate retraining cost ($\rho = 1$) a fixed calendar is the best policy in 5 of 8 experiments (with the period chosen in hindsight, which flatters it). Monitoring helps on Freddie Mac: retraining on alarms from recently arrived labels wins on freddie_2007, and on CBPE alarms on freddie_2016. Retraining on input-drift alarms is equivalent to always retraining, since those tests always fire: cheapest when retraining is free, among the worst once it costs anything. On tabred_ecom retraining never pays: newer models train on too little data.

7 Limitations

Simulated delays on six of eight experiments are constant; only Freddie Mac has natural arrival. Estimation results cover five seeds (model randomness and, for sampled streams, the sample); detection and retraining results are single runs with period-level summaries. The fixed-calendar policy’s period is chosen in hindsight. Regression tasks (five TabReD datasets) are not yet supported. Freddie Mac and TabReD data cannot be redistributed; the benchmark ships loaders and checksums, not data.

Freddie Mac, model built Jan 2007: estimated vs true default rate

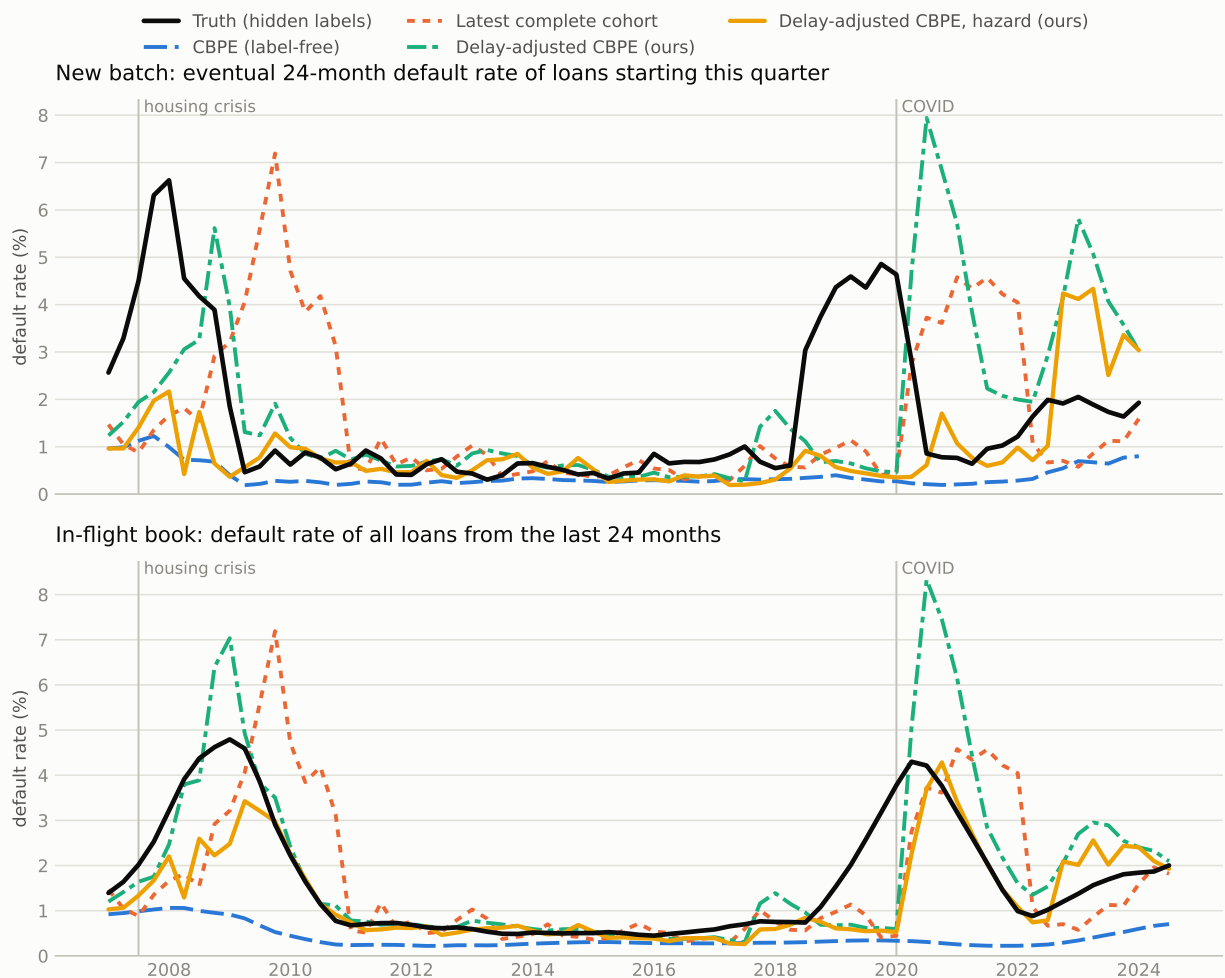


Figure 1: Freddie Mac, model built January 2007. Top: eventual default rate of each quarter’s new loans. Bottom: default rate of the in-flight book (loans from the last 24 months).

8 Conclusion

Monitoring benchmarks should model when labels arrive. Doing so shows that input-drift alarms are uninformative on real data, that label-free and label-based estimators fail in opposite directions, and that early-arriving labels, once their bias is corrected, close much of the gap.

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